

Suraj Krishna Banavalikar

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Skills

Programming Languages: Java, C/C++, Python, JavaScript

Core Computer Science: Data Structures & Algorithms, OOP, OS, DBMS, Computer Networks

Databases: PostgreSQL, MySQL, MongoDB, Pinecone

Technologies & Tools: Git, GitHub, Linux, Docker, Postman, Power BI

Frontend Development & Backend Systems: HTML, CSS, React.js, Node.js, Express.js, REST APIs

Education

Ramaiah Institute of Technology (MSRIT), Bengaluru
B.E. Computer Science and Engineering

Oct 2023 – Present
CGPA: 9.19/10

St. Joseph's PU College, Karwar
Course in PCMC (Computer Science)

Aug 2021 – Mar 2023
Grade: 96.16%

Project Work

Campus Helpdesk – AI Powered Campus Helpdesk System (2025)

MERN Stack, Python, Pinecone, JavaScript

- Built an NLP-powered query system (vector search + hybrid re-ranking with Pinecone) improving answer accuracy and reducing manual query handling by ~40%.
- Optimized performance with MongoDB connection pooling + async job queues (20 jobs/batch, retry with backoff), reducing API blocking and improving throughput under concurrent load.
- Architected a scalable Node.js backend using PM2 clustering (10 workers) with load balancing, supporting ~50–100 concurrent users and stable low-latency responses.
- Source Code:** surajkb2005.github.io/Resume/pages/campus_helpdesk.html

StellarRoute (2025)

Python, Leaflet.js, NASA APIs, Redis

- Designed and implemented a GPS-denied navigation system using inertial navigation (IMU-based dead reckoning).
- Processed accelerometer and gyroscope data to estimate position in high-interference environments, such as solar storms, to calculate position without satellite dependency.
- Integrated NASA's NOVA API for live kp index values and proposed future integration with ISRO's Aditya-L1 real-time telemetry.
- Source Code:** StellarRoute **Live App:** stellar-route.me

Zero Server AI Privacy Gateway (2026)

JavaScript, IndexedDB

- Developed a privacy-preserving browser extension that intercepts ChatGPT/Claude prompts and sanitizes sensitive data locally before transmission.
- Built a multi-layer PII detection engine using C++ → WebAssembly, regex validation, and heuristic analysis to detect 19+ categories of sensitive information.
- Implemented IndexedDB-based placeholder vault and automatic response restoration using DOM MutationObserver, ensuring seamless user experience.
- Achieved real-time processing with ~2.86 ms median latency and 92% detection accuracy while maintaining a zero-server architecture.
- Source Code:** [Zero_Server_AI_Privacy_Gateway](#)

Work Experience

RIT S.T.A.R.D.U.S.T

Software Engineer – Avionics Member

May 2025 – Present

- Contributed to early-stage development of software modules for autonomous drone systems.
- Assisted in simulation and testing of navigation and control logic.

Awards and Certificates

- Secured 2nd Place at wHACKiest 2025 (24-hour Intra-college level hackathon) among 100+ teams. Built the StellarRoute project as part of a team, contributing as a backend engineer.